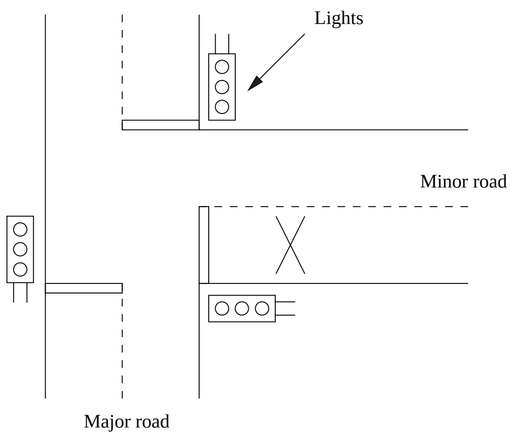
(c) Check whether *LightSwitch* is timelock-free and non-zeno.

EXERCISE 9.2. Consider the following two timed automata:

(a) (b)

As these timed automata have a single location only, the *state* of these timed automata can be considered as just a point in the real plane. A point (d, e) (with $d, e \geqslant 0$) thus represents that clock x has value d and clock y has value e. Determine the reachable state space of each of these timed automata. Justify your answers.

EXERCISE 9.3. (Modeling exercise). A control system must ensure the safe and correct functioning of a set of traffic lights at a T-junction between a major and a minor road. The lights will be set on green on the major road and red on the minor road unless a vehicle is detected by a sensor in the road just before the lights on the minor road. In this case the lights will be switchable in the standard manner and allow traffic to leave the minor road. After a suitable interval the lights will revert to their default position to allow traffic to flow on the major road again. Once a vehicle is detected the sensor will be disabled until the minor-road lights are set to red again. A sketch of the T-junction is provided below.



Questions:

1. First we ignore all timing issues involved and concentrate on the qualitative aspects of the behavior of the traffic lights. Model the above system as a network of (timed) automata. For convenience, you may assume that the two major-road lights are fully synchronized and can be modeled as a single light. Complement your system model by adding a process that regulates the arrival of cars in the minor road.

2. Adapt your model so as to incorporate the following timing constraints. Deal with each timing constraint separately so as reduce the complexity. Indicate for each timing constraint the necessary adaptations to your un-timed model:

   (a) a minor-road light stays on green for 30 seconds,

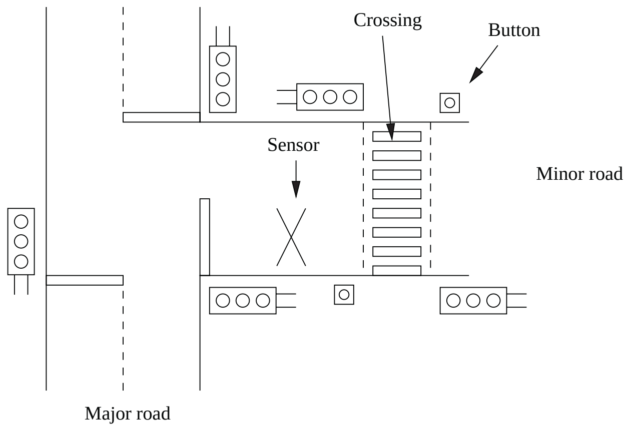
   (b) all interim lights stay on for 5 seconds,

   (c) there is a 1 second delay between switching one light off and another on (e.g., switching from red to yellow),

   (d) the major-road lights must be on green for at least 30 seconds in each cycle,

   (e) (more involved) but must respond to the sensor immediately after that.

3. We extend the T-junction in the following way. Suppose there is a pedestrian crossing a short distance down the minor road but beyond the sensor. There is a button on each side of the road for pedestrians to indicate they wish to cross. The crossing should only allow people to cross when the "minor lights" are set to red in order to minimize waiting times for traffic on the minor road. The new situation is sketched below.

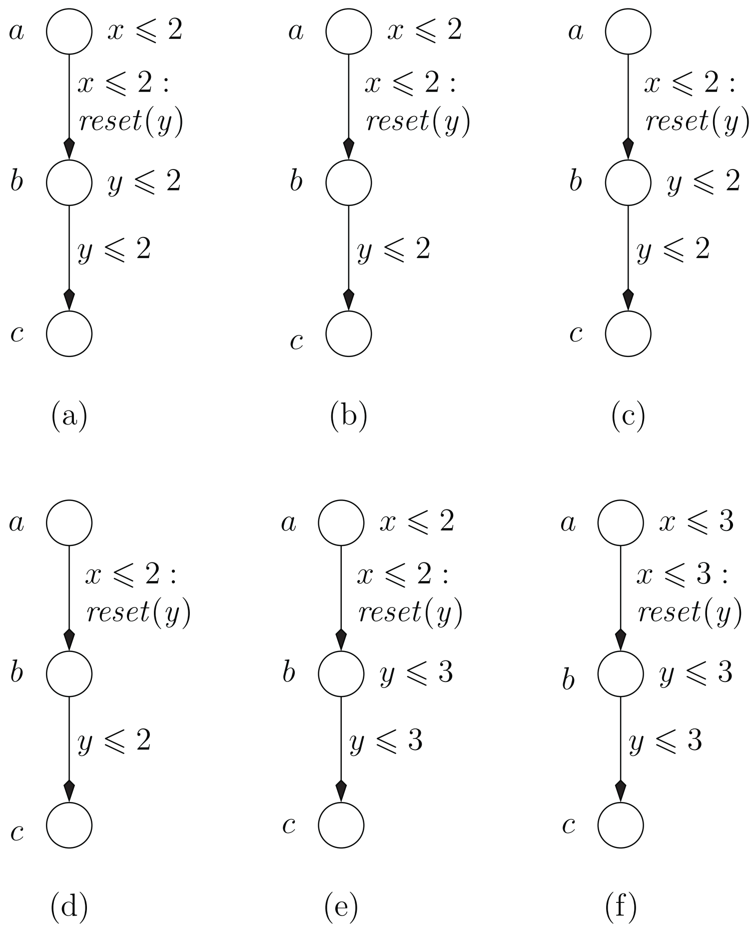


Extend your timed model of the previous question in order to cope with this new situation.

4. Does the crossing indeed only allow pedestrians to cross when the "minor lights" are set to red?

(This exercise is inspired by [151].)

Exercise 9.4. Consider the following six timed automata:



Give for each individual timed automaton a TCTL formula that distinguishes this timed automaton from all other ones. It is allowed to use only the atomic propositions a, b, and c and clock constraints in the TCTL formulae. Location identifiers are not allowed. (This exercise is due to Pedro D'Argenio.)